



Python Week 1

Introduction & Basics





Objective: Introduction to Python

What You'll Learn:

- What Python is and why people use it
- How to type and run Python code
- How to display text and input from users



A **computer** is a machine that:

- follows instructions
- cannot think on its own
- only does what it is told

If instructions are wrong → result is wrong



What is coding?

Coding is writing instructions for a computer.

These instructions tell the computer:

- what to do
- when to do it
- how to do it



Computers do NOT understand English.

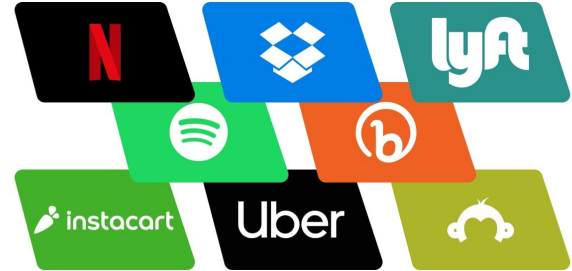
We use programming languages like **Python** to translate instructions.





Python is a beginner-friendly programming language.

- Used by companies, scientists, and programmers all over the world.
- Easy to read and write compared to other languages.





We will be using an online code editor called **Online Python**.

1. Go to the website <https://www.online-python.com> by searching up “Online Python” or via link in Google Classroom.
2. If there is text in the box, delete it.
3. Done! You are ready to start coding.



Basic “built-in” Functions



What is a variable?

A **variable** is a labeled container that stores data. You can store text, numbers, or values.

```
name = "Alex"
```

Think: name → label,
"Alex" → value



There are two main types of data:

1. Strings
2. Numbers



A **string** is anything inside quotation marks. This could be

- Text
- Letters
- Words

```
name = "Alex"  
hobby = "soccer"
```

Strings ALWAYS use quotes: " "



Try this:

Open Online Python and create a variable called snack which stores the name of your favorite snack.

The format should be:

```
[variable name] = "[String]"
```



A **number** is like a regular number found in your math class.

```
age = 13  
score = 100
```

Notice how numbers do NOT use quotes



Math in Python

```
print(5 + 3)    # addition  
print(10 - 4)   # subtraction  
print(6 * 2)    # multiplication  
print(8 / 2)    # division
```

Q: Can you guess what this prints out?

Copy into Online Python and click Run to check your answer!



Why this matters:

If you mix them up, the program will break :(

```
age = "13"    # string  
age = 13      # number
```

One of them is text, one is math



Experiment

Try typing both into Online Python:

```
print("5" + "5")
```

```
print(5 + 5)
```

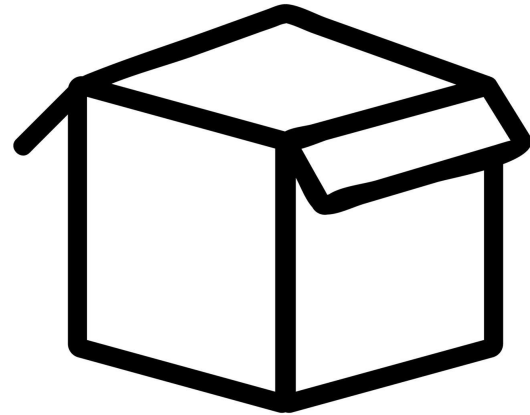
What happens when you run the code?



Why do variables matter?

Variables help us:

- Store information
- Reuse data
- Change values easily





Print Statements

print() shows output on the screen.

```
print("Hello World")
```

This tells the computer: "Show this on the screen!"



Putting it all together...

```
name = "Alex"  
print("My name is", name)
```

Q: Can you guess what that prints out?



To print multiple things in a sentence, use
commas!

What do the following print out?

a)

```
age = 13
print("I am", age, "years old")
```

b)

```
x = 5
y = 10
print("The total is", x + y)
```



Now it's your turn!

Open up Google Classroom and complete Week 1 Classwork.

Due Tomorrow

My First Python Program

Create a Python program that includes:

1. Three variables:

- name (string)
- age (number)
- favorite_subject (string)

2. Print all three variables in full sentences.

Challenge (Optional)

If you finish early:

- Add a 4th variable: favorite hobby
- Print a sentence about it

Upload a screenshot of your finished work to this assignment.





Now it's your turn!

Create a Python program that includes:

1. Three variables:
 - a. name (string)
 - b. age (number)
 - c. favorite_subject (string)
2. Print all three variables in full sentences.

challenge (optional)

If you finish early:

- a. Add a 4th variable: favorite hobby
- b. Print a sentence about it





Classwork Sample Answer

```
name = "Alex"  
age = 13  
favorite_subject = "math"  
  
print("My name is", name)  
print("I am", age, "years old")  
print("My favorite subject is", favorite_subject)
```



Thanks!

Have questions? Reach out:

cs4nyc0@gmail.com

cs4nyc0.wixsite.com/code4nyc



@code4nyc_



@code4nyc